When Health Data Do Not Connect

Maternal Continuity of Care, Data Fragmentation and the Case for Interoperable Health Information Systems in Nigeria

Data Sharing | Data Value Chain | Digital Public Infrastructure

Core policy proposition

Maternal continuity of care depends not only on whether services exist, but also on whether relevant information can follow women across the care pathway and be used effectively by institutions.

Opeyemi O. Agboola, MPH

Health Data Analytics and Public Health Policy

September 2026

Technical analysis and reproducible project materials: GitHub repository

1. Executive Summary

Nigeria has expanded access to maternal healthcare, but contact with individual services does not necessarily translate into continuous care throughout pregnancy, childbirth and the postnatal period. A woman may enter antenatal care and still fail to receive adequate antenatal contacts, skilled support at delivery, facility-based care or timely postnatal services. The policy challenge is therefore not only access. It is whether women remain connected to appropriate care across successive stages.

The survey-weighted analysis developed for this policy brief shows a steep decline across that pathway. Any antenatal care reached 73.65%, but coverage fell to 53.65% for four or more antenatal contacts, 45.09% for skilled birth attendance and 42.93% for facility delivery. Maternal and newborn postnatal care within two days were 42.87% and 42.27%, respectively. Only 24.52% achieved the complete continuum defined in the analysis. The largest early difference was between any antenatal care and four or more contacts, at about 20 percentage points.

Continuity is also unequal. Complete-continuum coverage differs substantially by education, household wealth, geographic zone, residence and age group. These disparities matter because a national average can improve while large groups remain disconnected from the care pathway.

This is also a data-governance problem. Maternal care often crosses facilities, levels of care and institutional systems. When information generated during antenatal care, referral, delivery and postnatal care is stored in disconnected systems, the health system may struggle to reconstruct a complete care pathway, identify missed transitions or target follow-up. The analysis does not claim that fragmented digital systems caused the observed service gaps. It shows why institutions need information systems capable of seeing those gaps clearly and responding to them.

Priority recommendations

- Make maternal continuity of care a priority national interoperability use case within Nigeria's digital health architecture.
- Define a minimum interoperable maternal health dataset with common definitions, identifiers and exchange standards.
- Strengthen patient identification, referral linkage and shared national registries while protecting privacy.
- Build clear institutional rules for responsible health data sharing, access, security and accountability.
- Shift maternal health monitoring from isolated service counts toward continuity-focused decision intelligence.

Main policy message

Nigeria should measure digital health success by whether information becomes more useful for care and decision-making, not simply by how many records or facilities become electronic.

2. Maternal Continuity of Care as a Data Problem

Maternal healthcare is a sequence of services rather than a single encounter. Antenatal care provides a platform for screening, prevention, diagnosis and preparation for childbirth, while postnatal care continues essential support for women and newborns after delivery (World Health Organization [WHO], 2016, 2022). A useful maternal pathway therefore links pregnancy, childbirth and early postnatal care rather than treating each stage as an isolated event.

Continuity can break for many reasons, including cost, distance, transportation, health literacy, quality of care, staffing constraints and weaknesses in referral systems. Digital systems do not remove these barriers. They can, however, determine how visible the pathway is to providers and decision-makers.

The distinction between service continuity and information continuity is important. Service continuity concerns whether a woman actually receives appropriate care across successive stages. Information continuity concerns whether relevant data generated during those stages remain available, interpretable and usable as care progresses. Weak information continuity does not prove weak service continuity, but it can make service gaps harder to identify and address.

Nigeria has formally recognised fragmentation in its health information environment. When the Federal Government inaugurated the implementation committee for the Nigeria Digital in Health Initiative in May 2024, the Federal Ministry of Health and Social Welfare described a system affected by non-uniform data collection, heavy reliance on paper records and electronic platforms that were not standardised to integrate and exchange patient information across institutions (Federal Ministry of Health and Social Welfare [FMoHSW], 2024).

Maternal healthcare makes this particularly consequential. A woman may receive antenatal care at a primary health centre, undergo investigations elsewhere, deliver in another facility and obtain postnatal care through a different service point. If the information generated at each stage remains isolated, every transition can become a new information gap.

A maternal health data value chain

> **Data generation -> storage -> sharing -> integration -> analysis -> decision-making**
>
> The public value of maternal health data depends on the entire chain. Collecting more data does not necessarily create more value if information cannot move, connect or influence action.

This framing links the maternal continuity problem to three fellowship themes directly: data sharing, the data value chain and Digital Public Infrastructure. The central question is not whether Nigeria needs one enormous database. It is whether different systems can retain their specialised purposes while exchanging selected information safely and consistently when clinical or policy needs require it.

3. Evidence from the Maternal Continuity-of-Care Analysis

The empirical analysis uses the 2024 Nigeria Demographic and Health Survey. Data collection took place from December 2023 to May 2024. The survey was implemented by the National Population Commission under the aegis of the Federal Ministry of Health and Social Welfare, with technical assistance from ICF through The DHS Program. The sample was designed to provide representative estimates nationally, for urban and rural areas, the six geopolitical zones, all 36 states and the Federal Capital Territory (FMoHSW, National Population Commission [NPC], & ICF, 2025).

The project applied survey weighting and developed national, equity and state-level estimates. It also included regression comparisons, sensitivity checks and a reproducible analytical workflow spanning R, SQL and reporting outputs. The figures below are project-derived estimates based on the analytical definitions used in this study. They should be read alongside, rather than as replacements for, official NDHS tabulations.

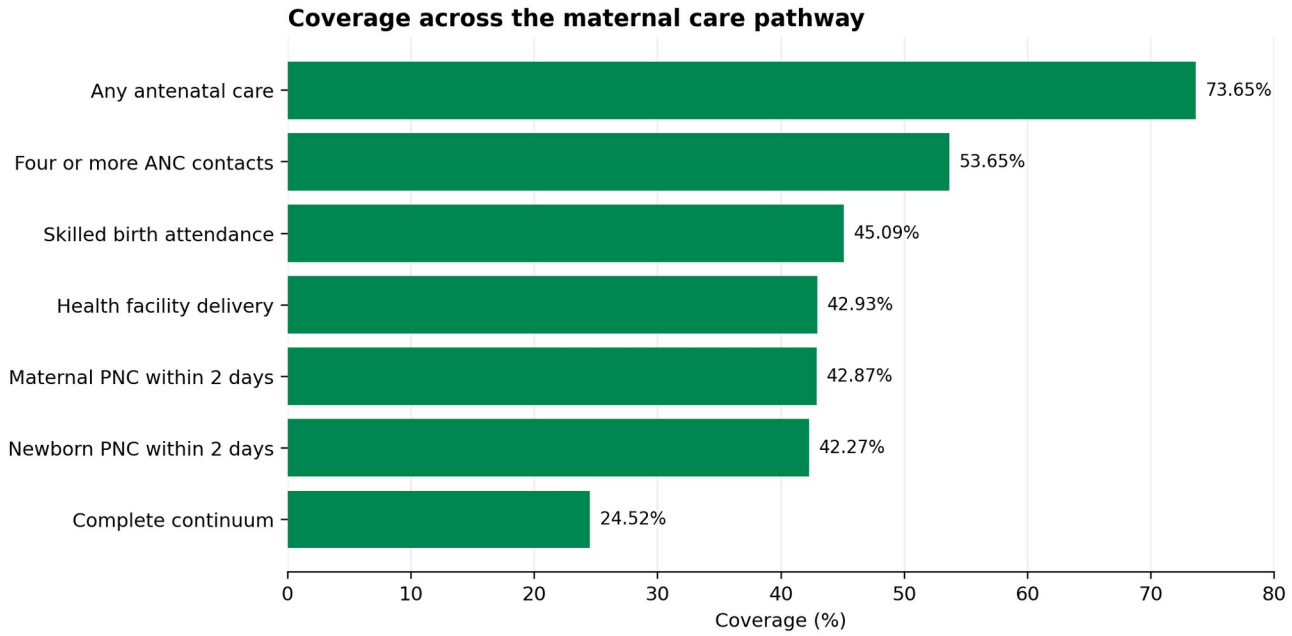


Figure 1. Coverage across the maternal care pathway. Source: author's survey-weighted analysis of the 2024 NDHS.

What the pathway shows

Entry into care is substantially higher than completion of the defined continuum. Any antenatal care was 73.65%, falling to 53.65% for four or more contacts and 45.09% for skilled birth attendance. Facility delivery and early postnatal care clustered around 42%, but only 24.52% achieved the complete continuum. This means that the policy story changes when services are examined together rather than one at a time.

The largest early difference was between any antenatal care and four or more ANC contacts, about 20 percentage points. Coverage then fell by about 8.56 percentage points between four or more ANC contacts and skilled birth attendance. A further large difference appears between newborn postnatal care and complete-continuum coverage. These are population-level differences between indicators, not proof that the same individual women dropped out at each stage.

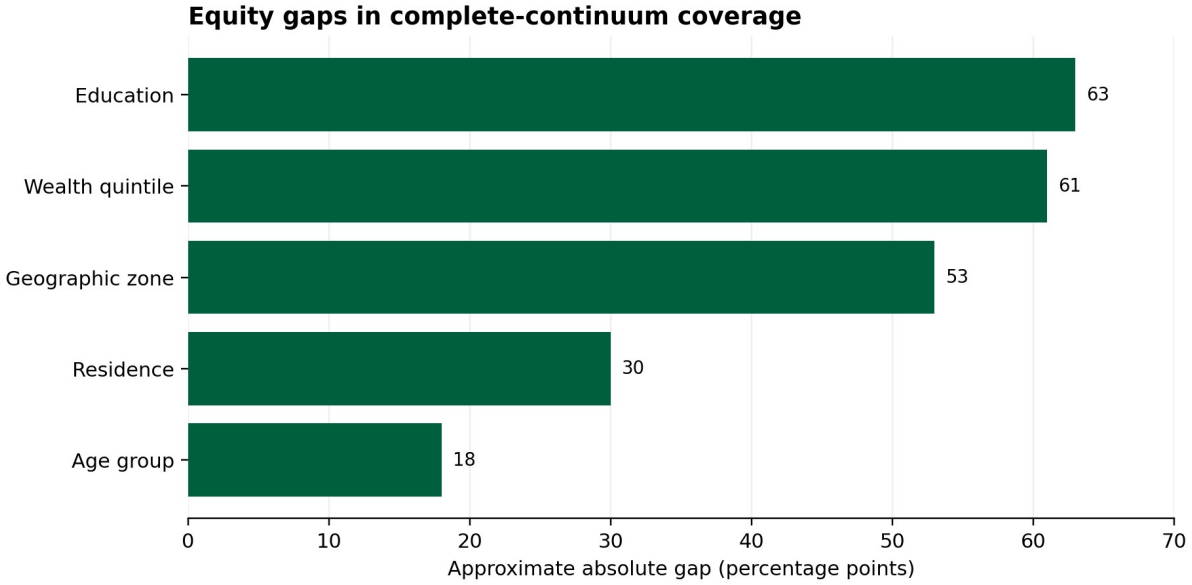


Figure 2. Approximate absolute equity gaps in complete-continuum coverage. Source: author's survey-weighted analysis of the 2024 NDHS.

Inequality is embedded within the continuity problem. Education and wealth produced some of the largest observed gaps, followed by geographic zone, residence and age group. For residence, complete-continuum coverage in the equity analysis was markedly higher among urban than rural women. State-level estimates also varied considerably, reinforcing the need for subnational rather than purely national monitoring.

Technical evidence

The public project repository contains the analytical workflow, R and SQL scripts, derived outputs and validation materials. Restricted DHS microdata are not redistributed. Repository: https://github.com/opeyemiagboolabethel/maternal-continuum-nigeria-2024

4. Where the Maternal Health Data Value Chain Breaks

The maternal health data value chain can weaken at several points. These weaknesses are cumulative: a problem created at the point of care can travel through storage, integration and analysis until it becomes a weaker decision.

Stage	Where value can be lost
Data generation	Incomplete records, inconsistent definitions, delayed entry and repeated documentation can reduce data quality at the point of care.
Storage	Paper registers and separate electronic platforms can leave antenatal, delivery, laboratory and postnatal information in different locations.
Sharing	Technical barriers, unclear responsibilities and weak governance can prevent relevant information reaching authorised users.
Integration	Different identifiers, formats, facility codes and terminology can stop shared data from being meaningfully linked.

Analysis	Routine reporting may count services without showing whether the same women progress across the pathway.
Decision-making	Dashboards and reports create little value when information is late, difficult to interpret or disconnected from operational responsibility.

The problem is not that every dataset must be merged. The problem is that important information can become stranded. A woman referred to another facility may arrive without a usable digital history. A programme may have separate antenatal and delivery counts but no reliable way to understand continuity. A state may receive several reports that require manual reconciliation before they can be used.

Nigeria's 2024 digital health reform explicitly identified these weaknesses, including fragmentation, lack of standardisation and limited real-time exchange (FMoHSW, 2024). The policy response should therefore focus on information continuity across the entire chain rather than simply increasing the volume of digital data collected.

A stronger chain would allow institutions to ask more useful questions: Where are women most likely to experience interruptions? Which facilities or communities require additional support? Are referral gaps visible? Which equity groups remain consistently underserved? Are improvements in reported coverage matched by improvements in continuity?

5. Data Sharing, Interoperability and Digital Public Infrastructure

Nigeria is already moving toward a more connected national architecture. In July 2025, the Federal Ministry of Health and Social Welfare described the Nigeria Digital Health Initiative as built around an interoperable digital health services network, a Health Claims Exchange and a Health Information Exchange intended to support secure sharing across public and private facilities (FMoHSW, 2025). In July 2026, the Ministry described the National Digital Health Architecture as an interoperable ecosystem anchored on the principle of "One Patient, One Health Record" (FMoHSW, 2026).

Common standards

Interoperability is more than connecting software. Systems must preserve the meaning of the information they exchange. WHO's 2021 Digital Adaptation Kit for Antenatal Care provides a useful model because it translates clinical recommendations into software-neutral data elements, workflows, decision support and indicator specifications designed for digital implementation (WHO, 2021). This allows different systems to follow common content rules without requiring one national software product.

Patient identification and shared registries

Continuity also depends on knowing whether records created at different places refer to the same woman. Nigeria's emerging national architecture includes registries for clients, facilities and health workers, alongside shared exchange infrastructure (FMoHSW, 2025). Used appropriately, such registries can reduce duplicate records, support referral linkage and make facility-level information more consistent across systems.

Responsible institutional data sharing

Stronger linkage increases the importance of governance. The Nigeria Data Protection Act 2023 classifies health status as sensitive personal data and creates obligations for lawful, secure and accountable processing (Nigeria Data Protection Commission [NDPC], 2023). Maternal health interoperability must therefore be designed around minimum necessary access, role-based permissions, authentication, audit trails, data-quality responsibilities and clear rules for correction, retention and breach response.

Digital Public Infrastructure

Digital Public Infrastructure offers a wider framework for these shared capabilities. The World Bank Group describes DPI as reusable digital building blocks developed for public benefit, including identity and trusted data sharing, with principles such as interoperability, privacy by design, inclusion, openness and strong governance (Clark et al., 2025). In health, this means individual applications can remain specialised while relying on shared rules and infrastructure to exchange information safely.

A practical maternal-health use case

Can an authorised provider access the relevant information needed when a woman moves from antenatal care to referral, delivery and postnatal care, while privacy is protected and the same information can also support continuity-focused monitoring?

6. Policy Recommendations

1. Make maternal continuity a national interoperability use case

What should change: Define maternal continuity as a priority implementation test for the National Digital Health Architecture. Specify the information that should move across antenatal care, referral, delivery and postnatal care, and use the pathway to test whether national exchange infrastructure is working in practice.

Who should act: Federal Ministry of Health and Social Welfare, State Ministries of Health, the National Primary Health Care Development Agency (NPHCDA), public and private facilities, digital health vendors and implementing partners.

Why it matters: Digital architecture becomes meaningful when it solves a real service problem. Maternal continuity provides a clear, measurable use case for Nigeria's emerging interoperable ecosystem (FMoHSW, 2025, 2026).

2. Establish a minimum interoperable maternal health dataset

What should change: Agree a national minimum set of maternal data elements covering identification, antenatal contacts, key risks, essential investigations, referral, delivery, maternal outcome, newborn information and early postnatal care. Align these elements with nationally adopted terminology and exchange standards.

Who should act: FMoHSW, digital health technical working groups, clinicians, midwives, health information managers, states, standards experts and vendors.

Why it matters: Systems cannot exchange meaningful information if the same concept is represented differently. WHO's Digital Adaptation Kit demonstrates how software-neutral content specifications can support consistent digital implementation (WHO, 2021).

3. Strengthen patient identification, referral linkage and shared registries

What should change: Use the national client, facility and health worker registries to improve matching across encounters and build referral linkage that can record whether a woman arrived, received care and completed follow-up.

Who should act: FMoHSW, State Ministries of Health, NPHCDA, facilities, relevant identity institutions and technology providers.

Why it matters: A woman should not become a disconnected record every time she changes facility. Better linkage supports both care coordination and continuity analysis, but must be implemented within the safeguards required for sensitive health data (NDPC, 2023).

4. Build responsible data sharing and privacy into implementation

What should change: Establish clear health-sector rules for role-based access, lawful purpose, minimum necessary data, authentication, audit logs, retention, correction, security incidents and institutional accountability.

Who should act: FMoHSW and NDPC, with state governments, public and private facilities, insurers, vendors and development partners.

Why it matters: Interoperability without trust is not sustainable. Nigeria needs connectivity and protection at the same time, especially where maternal and reproductive health information is involved (NDPC, 2023; Clark et al., 2025).

5. Shift maternal monitoring toward continuity-focused decision intelligence

What should change: Complement existing service indicators with routine views of progression across care stages, referral completion, state and facility variation, and equity by residence, education, wealth and geography.

Who should act: FMoHSW, State Ministries of Health, NPHCDA, monitoring and evaluation teams, facilities and research partners.

Why it matters: The analysis in this brief shows why isolated indicators can tell a less consequential story than the full pathway. Decision-makers need to see where continuity weakens, who is most affected and whether interventions are narrowing those gaps.

A realistic implementation sequence

- Define the maternal continuity use case and minimum dataset.
- Align selected systems and referral networks with shared standards and registries.
- Pilot secure exchange across a defined primary-to-secondary care pathway.
- Introduce continuity indicators into routine performance review.
- Evaluate effects on data quality, referral completion, targeting and decision-making before wider scale-up.

7. Conclusion

Nigeria's maternal continuity challenge is visible in the distance between initial contact with care and completion of the full pathway. In the analysis presented here, almost three quarters of women received some antenatal care, but only about one quarter achieved the defined complete continuum. The gaps were also socially and geographically unequal.

Those findings are primarily a service-delivery and equity concern. They are also a test of the information environment that policymakers need in order to respond. When care crosses facilities and systems, fragmented records make it harder to see where continuity weakens, connect referrals, compare performance and direct resources with precision.

Nigeria already has an important policy foundation through the Nigeria Digital in Health Initiative, the National Digital Health Architecture and emerging exchange and registry infrastructure. The next step is to make those investments prove their value against real care pathways.

Maternal continuity of care depends not only on whether services exist, but on whether information can follow women across the care pathway and be used effectively by institutions. A stronger interoperable health information environment will not solve every barrier to maternal care. It can, however, help Nigeria see those barriers more clearly, coordinate care more effectively and act on inequities sooner.

8. References

Clark, J., Marin, G., Ardic Alper, O. P., & Galicia Rabadan, G. A. (2025). Digital Public Infrastructure and Development: A World Bank Group Approach. Digital Transformation White Paper, Volume 1. World Bank Group. Source

Federal Ministry of Health and Social Welfare. (2024, May 31). FG inaugurates Committee on Digital in Health Initiative. Source

Federal Ministry of Health and Social Welfare. (2025, July 3). FG rallies stakeholders to build national digital health backbone. Source

Federal Ministry of Health and Social Welfare. (2026, July 30). FG reaffirms commitment to building an inclusive digital and resilient health system. Source

Federal Ministry of Health and Social Welfare of Nigeria, National Population Commission [Nigeria], & ICF. (2025). Nigeria Demographic and Health Survey 2024. Abuja, Nigeria, and Rockville, Maryland, USA: FMoHSW, NPC, and ICF. Source

Nigeria Data Protection Commission. (2023). Nigeria Data Protection Act 2023. Source

World Health Organization. (2016, November 28). WHO recommendations on antenatal care for a positive pregnancy experience. Source

World Health Organization. (2021, February 17). Digital Adaptation Kit for Antenatal Care: Operational requirements for implementing WHO recommendations in digital systems. Source

World Health Organization. (2022, March 30). WHO recommendations on maternal and newborn care for a positive postnatal experience. Source

Project repository

Agboola, O. O. (2026). Maternal Continuum of Care in Nigeria: survey-weighted analysis, SQL workflow and reproducible project materials. GitHub. https://github.com/opeyemiagboolabethel/maternal-continuum-nigeria-2024

Methodological note

The policy brief draws on a custom survey-weighted analysis of the 2024 NDHS microdata. The analytical definitions, code, validation outputs and derived public tables are documented in the linked repository. Restricted respondent-level DHS data are not included in the public repository.